

Phase 2 (Coral USB TPU) — Procurement Checklist

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A pragmatic shopping list to take your MVP into the AI phase on the **Raspberry Pi 4 + Coral USB Accelerator**. Items are grouped by priority. Pick the “Nice-to-have” based on budget and environment.

■ Must Have Hardware

- **Coral USB Accelerator (Edge TPU)** – USB 3.0 model.
- **Short, quality USB 3.0 cable** – 0.3–1 m; avoid long/cheap cables.
- **Raspberry Pi 4 power supply (official)** – stable 5V/3A.
- **MicroSD card 64–128 GB (U3/V30)** – room for footage + models.
- **Spare CSI ribbon cable (15–30 cm)** – for your Pi Camera Module 3 (backup).
- **Camera mount/arm** – rigid, non-slip; small tripod or clamp arm.
- **Cable management** – clips/ties to keep cables above water line.

■ Must Have Software/Deps (on the Pi)

- **EdgeTPU runtime**: ``libedgetpu1-std`` (or ``libedgetpu1-max`` for max clock)
- **TFLite runtime (aarch64)**: ``tflite-runtime``
- **PyCoral**: ``pycoral``
- **OpenCV (Python)**: ``python3-opencv``
- **InfluxDB token & bucket** – (org: ``home``, bucket: ``fish``)
- **Grafana access** – confirm admin creds

> Install commands (recap):

...

```
sudo apt update
```

```
sudo apt install -y libedgetpu1-std python3-pip python3-opencv v4l-utils
```

```
python3 -m pip install --upgrade pip
```

```
python3 -m pip install --upgrade tflite-runtime==2.14.0 pycoral==2.0.0
```

...

■ Model & Data (choose your path)

- **Zero training start**: **EdgeTPU friendly detector** (e.g., **EfficientDet-Lite0 INT8**).
- **Light fine tune (recommended)**:
 - **Labeling tool**: Roboflow or Label Studio (free tier is fine).
 - **~300–800 images** from your tank (day/night variety).
 - **Training env**: local PC or cloud (Colab/Kaggle/GCP).

■ Nice to Have Hardware

- **Powered USB 3.0 hub** (if other USB devices share the bus).
- **Small fan / heatsink for Pi** (if enclosure traps heat).
- **LED light strip (neutral/6500K)** for consistent lighting.
- **Black foam board / matte backdrop** to reduce reflections behind tank.
- **UPS or surge protector** (GFCI outlet, drip loop for safety).
- **External SSD (USB 3)** if you plan to record lots of video for training.
- **Water resistant enclosure or splash guard** for Pi and power connections.

■ Setup/Service Quality of Life

- **MicroSD card reader** (fast UHS-II preferred for flashing).
- **Spare microSD (32 GB)** as a rollback image.
- **ESD strap, precision screwdriver**, microfiber cloth & lens wipes.

■ Calibration Aids

- **Ruler or checkerboard print** taped behind the tank edge → lets you convert pixels to cm.
- **Masking tape** to mark fixed camera position/angle once dialed in.

■ Credentials & Config

- InfluxDB: URL, **org**, **bucket**, **token** stored in ``.env``.
- Grafana: create a **datasource** for Influx + a dashboard folder ("Fish Monitor").
- Git repo (local or private remote) for your code + configs.

Minimal Bill of Materials (quick view)

- Coral USB Accelerator (1)
- USB 3.0 cable, short (1)
- microSD 64-128 GB U3 (1)
- Spare CSI ribbon 15-30 cm (1)
- Camera mount/clamp/tripod (1)
- Cable clips/ties (assorted)
- (Optional) LED strip, powered USB hub, fan/heatsink, external SSD, UPS

Next Steps After Delivery

1. Plug Coral into a **blue USB 3.0** port → run the smoke test:
 - ``lsusb | grep -i google`` and `PyCoral `list_edge_tpus()``
 2. Place/mount the camera, set consistent lighting.
 3. Drop the compiled **_edgetpu.tflite** model into `~/fish-monitor/models/``.
 4. Run the Coral tracking script and confirm **fish_count** ≈ 7 .
 5. Start sending **distance_px/min** → Influx and view in Grafana.
- You now have everything needed to complete Phase 2 with Coral.**